Copyleaks

Scan details

Scan Time	Total Pages	Total Words
September 21st, 2025 at 09:50 UTC	3	528

Analysis Report

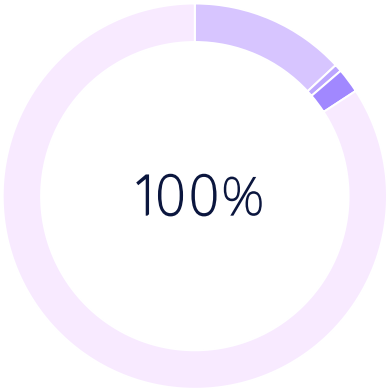
AI Detection Report

New Scans 9:50 AM

Plagiarism Detection

Plagiarism Types	Text Coverage
Identical	
Minor Changes	
Paraphrased	
Excluded	
Omitted Words	

AI Detection



	Text Coverage	Words
AI Text	100%	481
Low Frequency		63
Medium Frequency		3
High Frequency		10
Human Text	0%	0
Excluded		
Omitted Words		47

AI Content

	Text Coverage	Words
AI Text	100%	481
Low Frequency		63
Medium Frequency		3
High Frequency		10
Human Text	0%	0
Excluded		
Omitted Words		47

About AI Detection

Our AI Detector is the only enterprise-level solution that can verify if the content was written by a human or generated by AI, including source code and text that has been plagiarized or modified. Learn more

AI Text

A body of text that has been generated or altered by AI technology. Learn more

Human Text

Any text that has been fully written by a human and has not been altered or generated by AI. Learn more

Copyleaks AI Detector Effectiveness

Credible data at scale, coupled with machine learning and widespread adoption, allows us to continually refine and improve our ability to understand complex text patterns, resulting in over 99% accuracy–far higher than any other AI detector–and improving daily. Learn more

Ideal Text Length

The higher the character count, the easier for our technology to determine irregular patterns, which results in a higher confidence rating for AI detection. Learn more

Reasons It Might Be AI When You Think It's Not

The AI Detector can detect a variety of AI-generated text, including tools that use AI technology to paraphrase content, auto-complete sentences, and more. Learn more

User AI Alert History

Historical data of how many times a user has been flagged for potentially having AI text within their content. Learn more

AI Logic

The number of times a phrase was found more frequently in AI vs human text is shown according to low, medium, and high frequency. Learn more

AI Logic

Shows you the "why" behind AI detection with sources you can see and verify.

AI Phrases

Detects phrases that appear with higher frequency in AI-written text than in human writing.

The frequency of a phrase in AI vs. human text.

3 x — 162x

162x **it against existing methods.**

How frequently the phrase was found in our dataset:

AI Text	**1.91 / 1,000,000 Documents**
Human Text	**0.01 / 1,000,000 Documents**

134x **Common approaches include**

How frequently the phrase was found in our dataset:

AI Text	**124.01 / 1,000,000 Documents**
Human Text	**0.93 / 1,000,000 Documents**

114x **and AI safety**

How frequently the phrase was found in our dataset:

AI Text	**26.74 / 1,000,000 Documents**
Human Text	**0.23 / 1,000,000 Documents**

72x **propagation is central**

How frequently the phrase was found in our dataset:

AI Text	**1.13 / 1,000,000 Documents**
Human Text	**0.02 / 1,000,000 Documents**

48x **Since u1+u2≥u12+u22u_1+u_2 \geq \sqrt{u_1^2+u_2^2}u1+u2≥u12+u22, the**

How frequently the phrase was found in our dataset:

AI Text	**19.74 / 1,000,000 Documents**
Human Text	**0.41 / 1,000,000 Documents**

37x **Benchmarks demonstrate that:**

How frequently the phrase was found in our dataset:

AI Text	**45.08 / 1,000,000 Documents**
Human Text	**1.21 / 1,000,000 Documents**

32x **represented as an ordered pair**

How frequently the phrase was found in our dataset:

AI Text	**4.25 / 1,000,000 Documents**
Human Text	**0.13 / 1,000,000 Documents**

29x **represented as ordered pairs**

How frequently the phrase was found in our dataset:

AI Text	**3.84 / 1,000,000 Documents**
Human Text	**0.13 / 1,000,000 Documents**

24x **If u1,u2≥0u_1,u_2 \geq 0u1,u2≥0, then uout≥max(u1,u2)u_{out} \geq**

How frequently the phrase was found in our dataset:

AI Text	**9.22 / 1,000,000 Documents**
Human Text	**0.38 / 1,000,000 Documents**

23x **accuracy, computational cost, and**

How frequently the phrase was found in our dataset:

AI Text	**2.55 / 1,000,000 Documents**
Human Text	**0.11 / 1,000,000 Documents**

20x **Framework for Uncertainty**

How frequently the phrase was found in our dataset:

AI Text	**13.46 / 1,000,000 Documents**
Human Text	**0.67 / 1,000,000 Documents**

15x **deterministic and reproducible.**

How frequently the phrase was found in our dataset:

AI Text	**2.75 / 1,000,000 Documents**
Human Text	**0.18 / 1,000,000 Documents**

11x **Interval Arithmetic Interval**

How frequently the phrase was found in our dataset:

AI Text	**1.61 / 1,000,000 Documents**
Human Text	**0.15 / 1,000,000 Documents**

5x **its algebraic properties,**

How frequently the phrase was found in our dataset:

AI Text	**2.24 / 1,000,000 Documents**
Human Text	**0.44 / 1,000,000 Documents**

4x **addition and multiplication and**

How frequently the phrase was found in our dataset:

AI Text	**9.65 / 1,000,000 Documents**
Human Text	**2.23 / 1,000,000 Documents**

4x **framework with clear**

How frequently the phrase was found in our dataset:

AI Text	**2.7 / 1,000,000 Documents**
Human Text	**0.65 / 1,000,000 Documents**

4x **and interval arithmetic**

How frequently the phrase was found in our dataset:

AI Text	**1.59 / 1,000,000 Documents**
Human Text	**0.41 / 1,000,000 Documents**

4x **It provides reproducible**

How frequently the phrase was found in our dataset:

AI Text	**1.18 / 1,000,000 Documents**
Human Text	**0.33 / 1,000,000 Documents**

3x **Monte Carlo sampling [3], and**

How frequently the phrase was found in our dataset:

AI Text	**4.78 / 1,000,000 Documents**
Human Text	**1.45 / 1,000,000 Documents**

3x **Monte Carlo sampling [3], and**

How frequently the phrase was found in our dataset:

AI Text	**4.78 / 1,000,000 Documents**
Human Text	**1.45 / 1,000,000 Documents**

N/U Algebra: A Conservative Linear Framework for Uncertainty Propagation

Abstract

Uncertainty propagation is central to scientific inference, experimental design, engineering tolerances, and reproducible go/no-go decisions. Common approaches include Gaussian error propagation [1], Monte Carlo sampling [3], and interval arithmetic [2]. Each makes trade-offs between accuracy, computational cost, and reproducibility. This paper introduces N/U algebra, a linear framework with clear semantics: quantities are represented as ordered pairs (n,u)(n,u)(n,u) with u≥0u \geq 0u≥0, uncertainty propagates without quadratic cross-terms, and a conservation law is maintained. We prove algebraic properties of N/U addition and multiplication and establish a conservativeness theorem: N/U always produces equal or larger uncertainty than Gaussian error propagation. Benchmarking against Monte Carlo and interval arithmetic shows that (1) N/U yields consistently larger estimates than Gaussian and Monte Carlo, and (2) N/U is tighter than interval arithmetic. These properties make N/U a reproducible, assumption-light alternative for early-stage scientific and engineering analysis when setting go/no-go bounds.

1. Introduction

Uncertainty quantification underpins reproducible science, engineering tolerances, and go/no-go decision-making. Standard methods are Gaussian error propagation [1], Monte Carlo sampling [3], and interval arithmetic [2]. Each is limited by assumptions, expense, or pessimism.

This paper defines N/U algebra as a linear alternative. We demonstrate its algebraic properties, prove its conservativeness, and benchmark it against existing methods.

2. Methods: Definition of N/U Algebra

2.1 Representation

A physical quantity is represented as an ordered pair (n,u)(n,u)(n,u), with u≥0u \geq 0u≥0.

2.2 Operators

Addition:

(n1,u1)⊕(n2,u2)=(n1+n2, u1+u2)(n_1,u_1) \oplus (n_2,u_2) = (n_1+n_2, \; u_1+u_2)(n1,u1)⊕(n2,u2)=(n1+n2,u1+u2)

Multiplication:

(n1,u1)⊗(n2,u2)=(n1n2, n1u2+n2u1)(n_1,u_1) \otimes (n_2,u_2) = (n_1n_2, \; n_1u_2+n_2u_1)(n1,u1)⊗(n2,u2)=(n1n2,n1u2+n2u1)

Quadratic terms u1u2u_1u_2u1u2 are excluded.

2.3 Conservation

Define magnitude M=|n|+uM = |n|+uM=|n|+u.

Catch operator:

Cα(n,u)=(0,|n|+u)C_\alpha(n,u) = (0, |n|+u)Cα(n,u)=(0,|n|+u)

3. Properties and Proofs

3.1 Commutative and Associative

⊕ and ⊗ are commutative and associative by symmetry.

3.2 Monotonicity

If u1,u2≥0u_1,u_2 \geq 0u1,u2≥0, then uout≥max(u1,u2)u_{out} \geq \max(u_1,u_2)uout≥max(u1,u2).

3.3 Conservativeness

Theorem: For independent variables, N/U uncertainty is always ≥ Gaussian.

Proof: Gaussian propagation gives u12+u22\sqrt{u_1^2+u_2^2}u12+u22. N/U gives u1+u2u_1+u_2u1+u2. Since u1+u2≥u12+u22u_1+u_2 \geq \sqrt{u_1^2+u_2^2}u1+u2≥u12+u22, the result follows.

3.4 Relation to Interval Arithmetic

Interval multiplication yields |a|u2+|b|u1+u1u2|a|u_2+|b|u_1+u_1u_2|a|u2+|b|u1+u1u2. N/U omits u1u2u_1u_2u1u2, hence uNU≤uIntervalu_{NU} \leq u_{Interval}uNU≤uInterval.

4. Results

4.1 Numerical Comparison

Table 1. Uncertainty propagation under different methods.

CaseOperationOperandsN/U uGaussian uMonte Carlo (N=5000)Interval u1Add10±1 ⊕ 5±0.51.51.1181.121.52Multiply10±1 ⊗ 2±0.220±420±2.8320±2.820±4.23Multiply0.8±0.05 ⊗ 0.8±0.050.64±0.080.64±0.0560.64±0.0570.64±0.082

4.2 Hardy-Weinberg Example

Counts: A1A1=108, A1A2=74, A2A2=53 (N=235). Allele frequency: p=0.6±0.01p=0.6 \pm 0.01p=0.6±0.01.

Expected p2=0.36p^2 = 0.36p2=0.36.

N/U propagation: 0.36±0.0120.36 \pm 0.0120.36±0.012.

Gaussian propagation: 0.36±0.0080.36 \pm 0.0080.36±0.008.

N/U bands are wider, demonstrating conservativeness, yet remain deterministic and reproducible.

5. Discussion

The N/U framework is deterministic, conservative, and assumption-light. It provides reproducible first-pass bounds for experimental design, engineering tolerances, and AI safety envelopes. Its main limitation is over-conservatism and lack of distributional modeling.

6. Conclusions

N/U algebra is a linear uncertainty framework. Benchmarks demonstrate that:

N/U consistently produces larger uncertainty estimates than Gaussian error propagation or Monte Carlo.

N/U is tighter than interval arithmetic.

These properties make N/U algebra a practical tool for reproducible first-pass go/no-go decision-making.

References

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